



UNIVERSITAT POLITÈCNICA DE CATALUNYA  
BARCELONATECH

Centre de la Imatge i la Tecnologia Multimèdia

# Introduction

## 1.1 General aspects

Sergi Abadal (abadal@ac.upc.edu)

Xarxes i Jocs On-line – 2025-2026  
Grau en Disseny i Desenvolupament de Videojocs

## **Theory:**

Sergi Abadal ([abadal@ac.upc.edu](mailto:abadal@ac.upc.edu))

Campus Nord UPC, C6-112, Barcelona

Questions: after class or appointments via email

## **Lab sessions:**

Laura Gotarra (Terrassa) and Manel Rello (Barcelona)

## **Who are you?**

What do you know about networks?

What do you know about on-line game programming?

What do you know about their interrelation?

What do you want to obtain from this subject?



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# Course objectives and syllabus

# Anatomy of a headshot



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# The invariant of on-line games



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USER

GAME

NETW



Player 1 moves and shoots  
Player 1 changes state



Internet



# The invariant of on-line games

USER

GAME

NETW



Player 1 moves and shoots  
Player 1 changes state

Player 2 receives info  
Player 2 changes state



## THE SEAMLESS ILLUSION

We want the actions from player 1  
and its consequences to be seen  
equally in all players

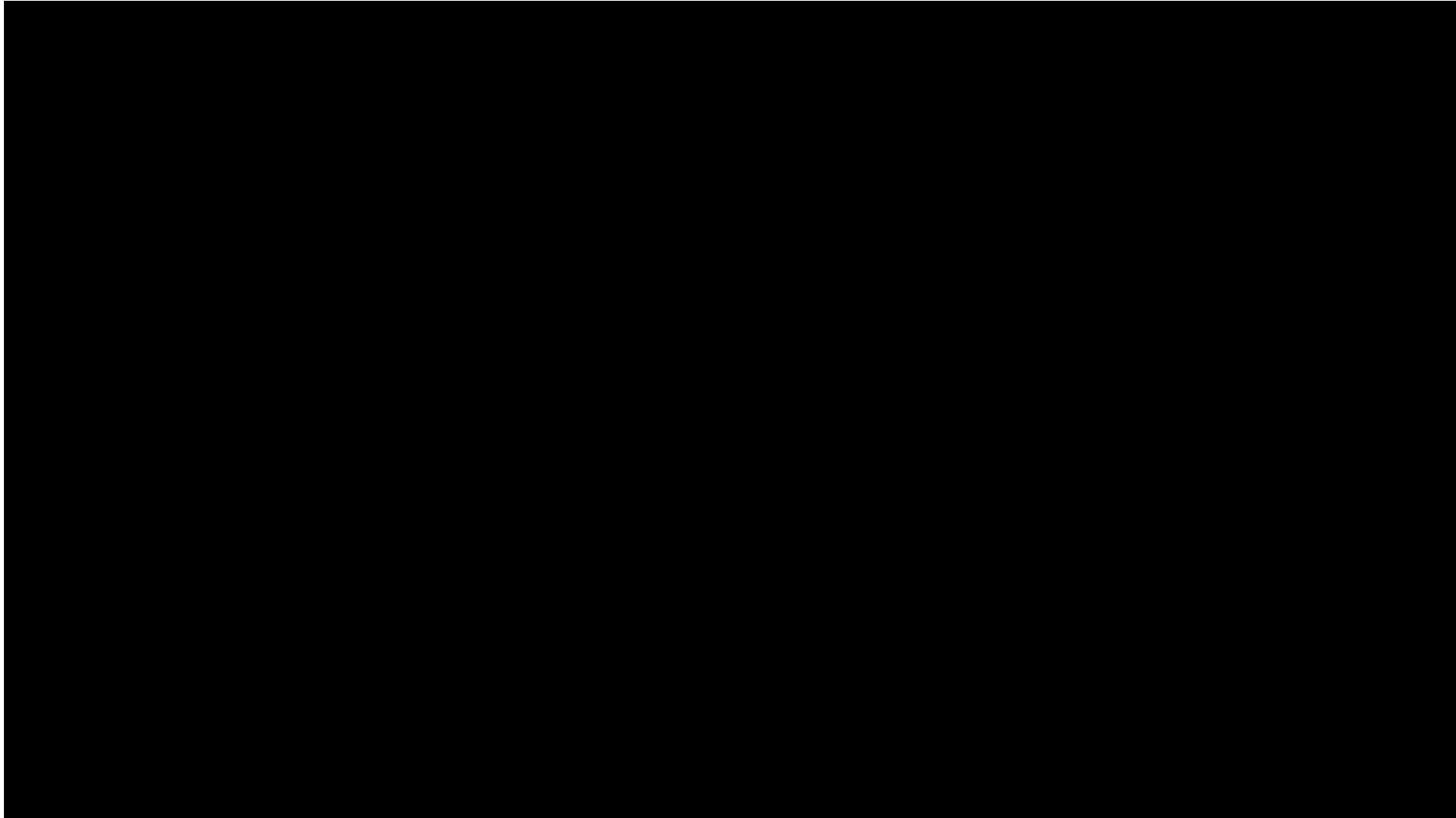


Internet

# The invariant of on-line games

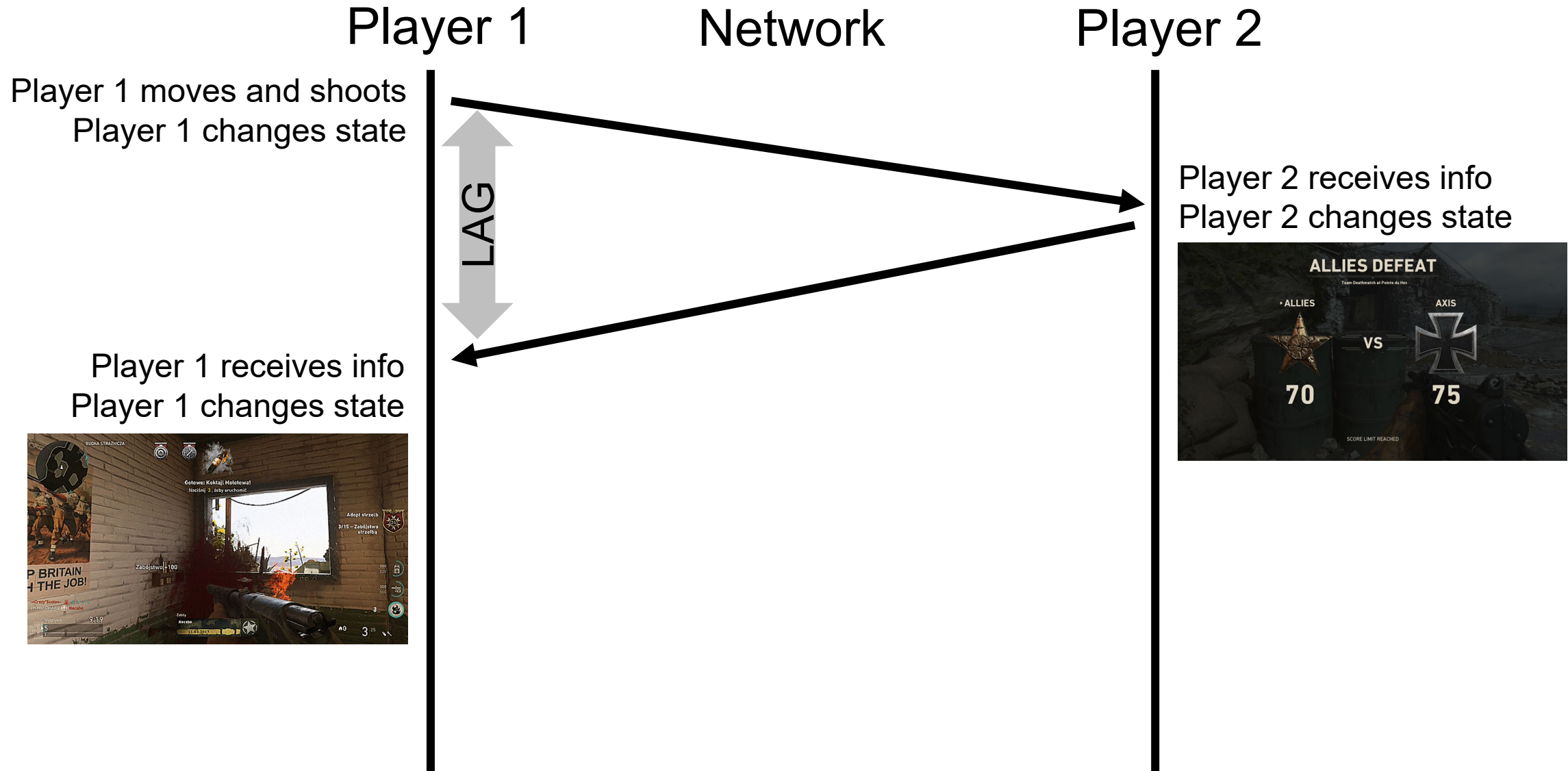


CITM





# Ideal time diagram of a headshot



# Problems?



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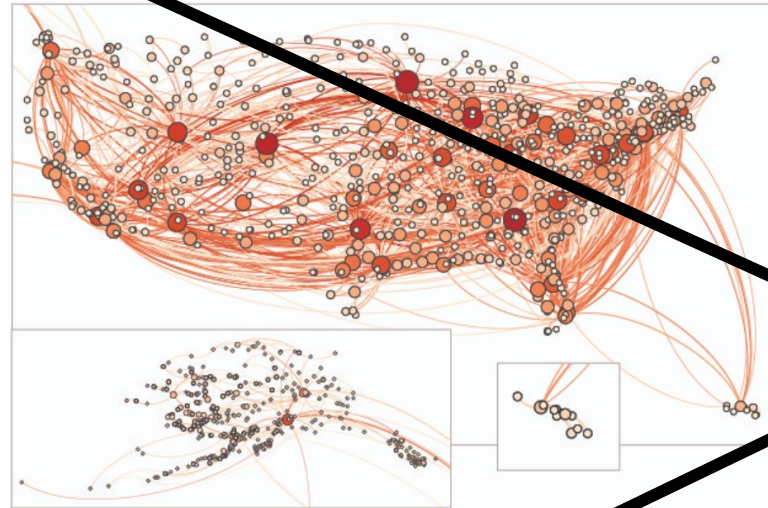
Player 1

Network

Player 2

Player 1 moves and shoots  
Player 1 changes state

LAG



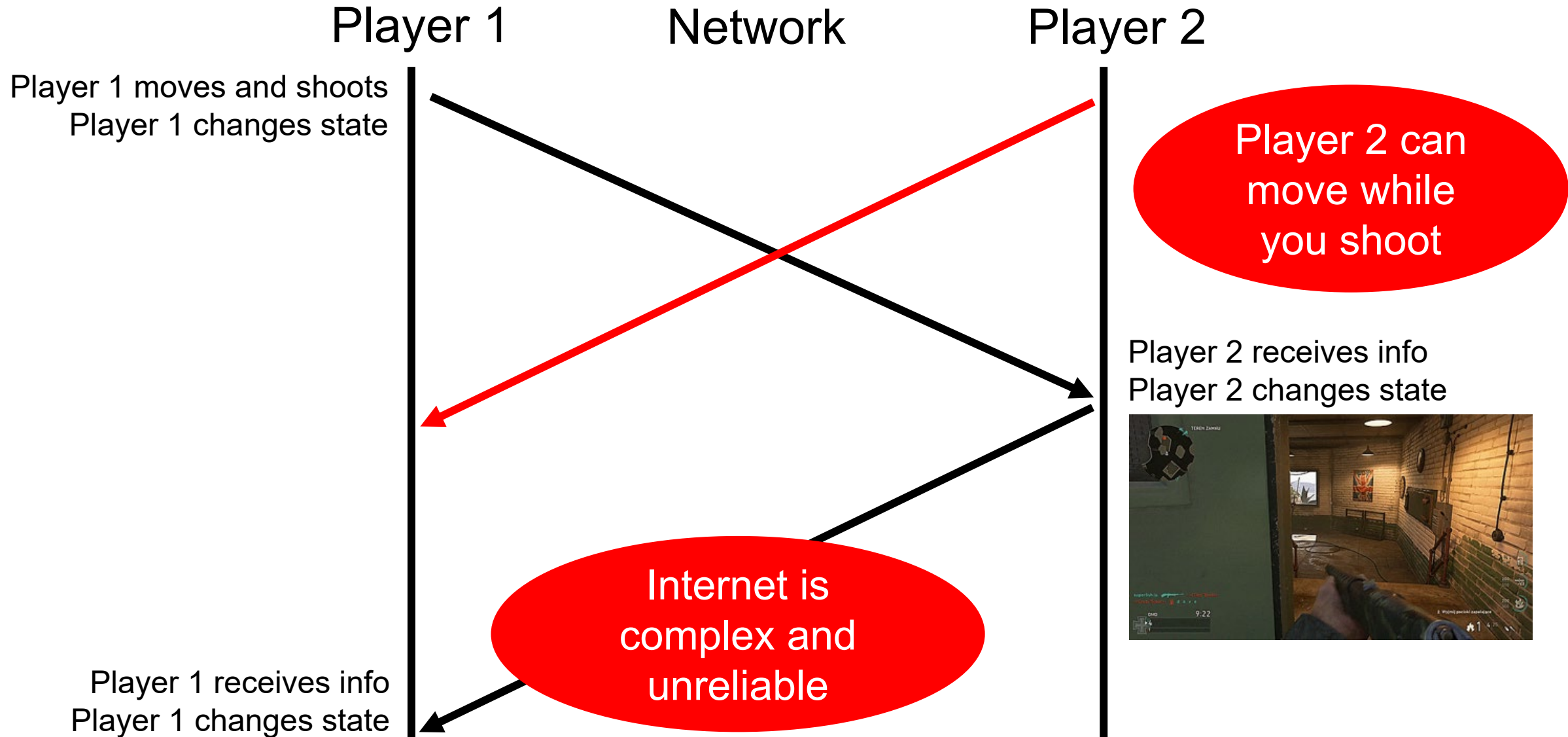
Player 2 receives info  
Player 2 changes state



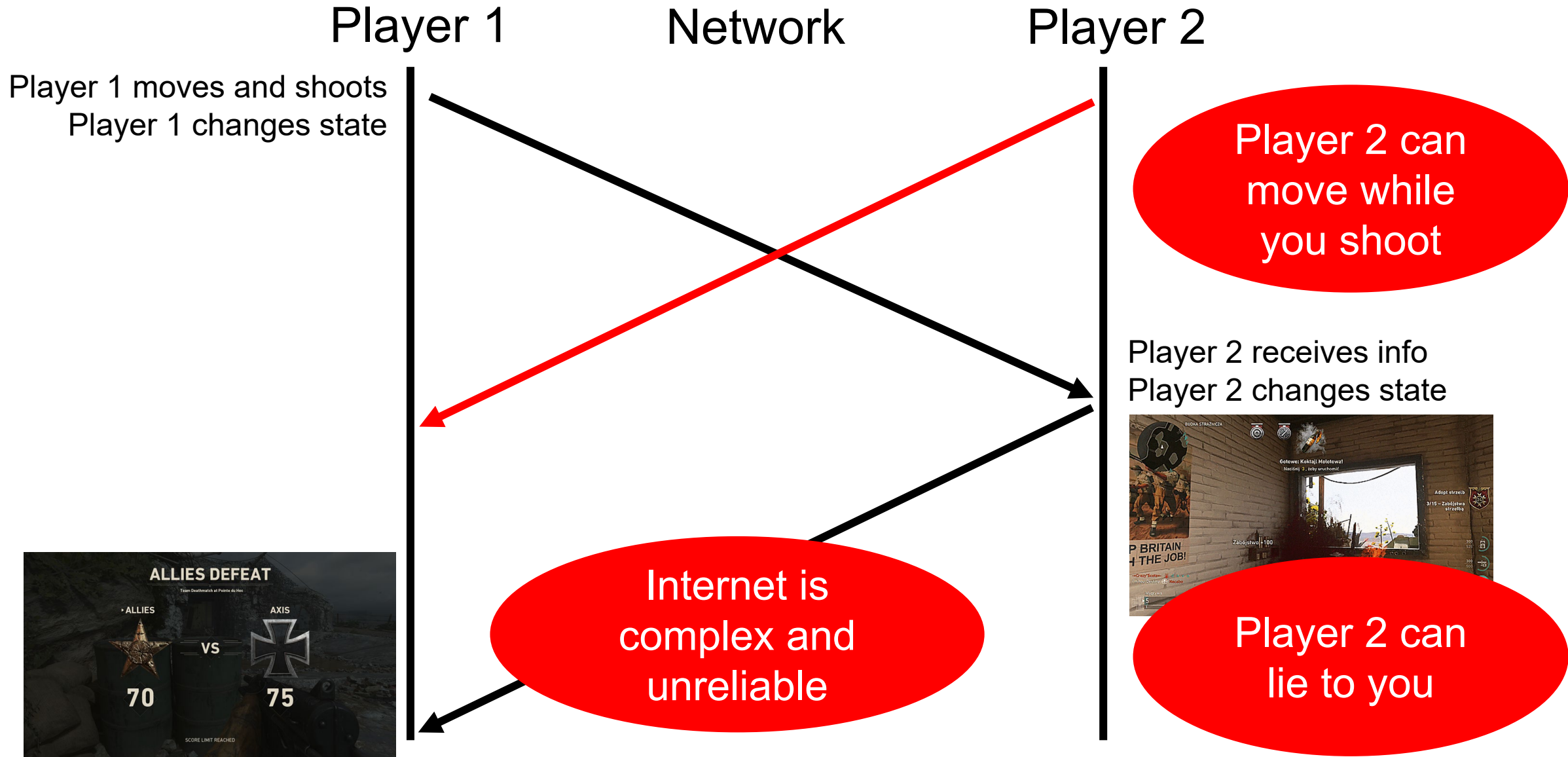
Player 1 receives info  
Player 1 changes state



# Problems?



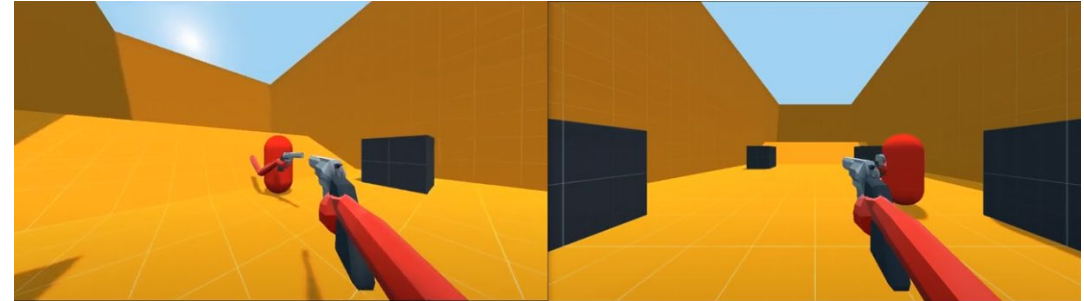
# Problems?





# Challenges of on-line games

1. Create the seamless illusion (synchronization model)
2. Create this illusion efficiently (object replication strategy)
3. Be resilient to network problems (fault tolerance, latency hiding)
4. Be compatible to multiple platforms (data serialization)
5. Reach to many simultaneous users (scalability)



# Example: Age of Empires

- Peer to peer architecture
- Deterministic lockstep model
- Send commands, not individual orders
- Filters to avoid communication bursts





# Example: Starsiege Tribes

- Client-server architecture
- State synchronization model
- Send game state to be confirmed by the server
- Differentiated code for the possible things happening in the game that depend on the network
  - Movements
  - Shots
  - Objects



# Course objectives



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1. Learn enough theory about computer networking to understand and quantify the problems that on-line games will suffer.
2. Gain insight on the impact of networking problems in on-line games and how to design them efficiently to address these problems.
3. Acquire hands-on experience on the programming of *netcode*, the part of a game's code that interfaces with the network and handles its problems.

## **1. Introduction (2 sessions)**

- General aspects
- Fundamentals of networking
- Graph theory

## 2. The protocol stack

## 3. Impact on programming of on-line games

## 4. Other stories

## 1. Introduction

## **2. The protocol stack (4 sessions)**

- Introduction and upper layers
- Lower layers: network, link, physical
- Other relevant protocols
- Putting it all together: Wireshark

## 3. Impact on programming of on-line games

## 4. Other stories

1. Introduction
2. The protocol stack
- 3. Impact on programming of on-line games  
(4 sessions)**
  - Summary and general aspects
  - Latency, jitter, and reliability
  - Latency hiding techniques
  - Scalability
4. Other stories



1. Introduction
2. The protocol stack
3. Impact on programming of on-line games
- 4. Other stories (2 sessions)**
  - Theory presentations

## LAB SESSIONS

1. Programming with threads (1 session)
2. Sockets (2 sessions)
3. Data serialization (2-3 sessions)
4. World state replication (2-3 sessions)
5. Latency and jitter mitigation (2-3 sessions)
6. Complete online game (2-3 sessions)

More details in  
the lab sessions

- Mid-term exam: 15%
- Theory presentation: 10%
- Final exam: 25%
- Lab sessions: 50%

REAVA covers mid-term  
and final exams (40%)

# Grades: Lab Sessions (50%)

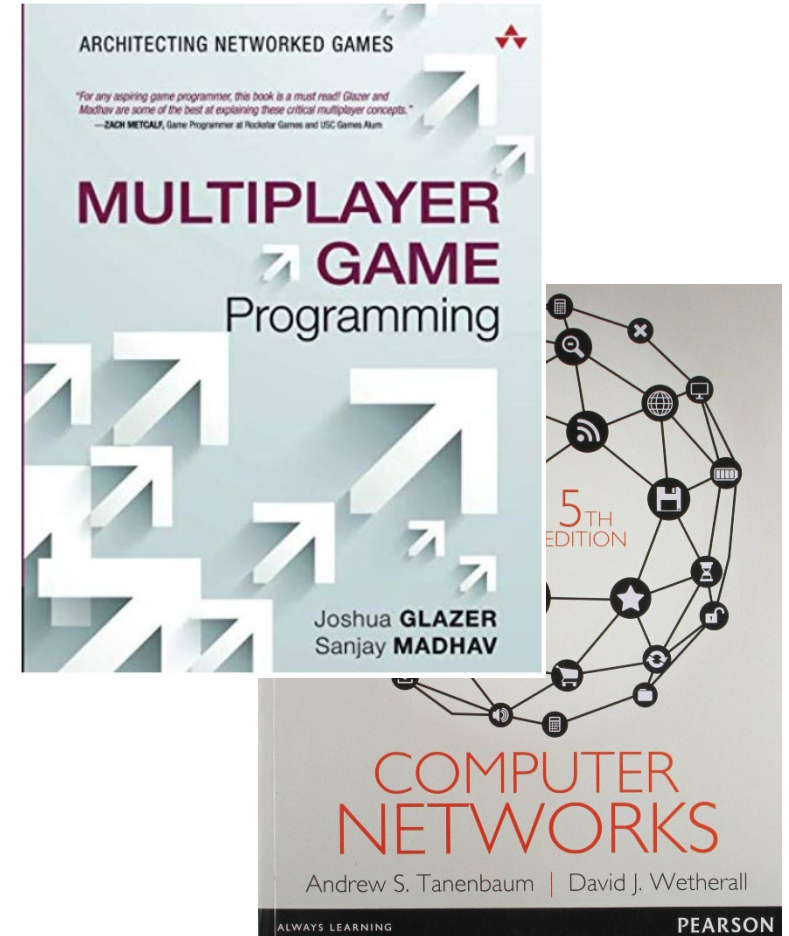


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- Threads: 0%
- Sockets: 5%
- Serialization: 15%
  - First deliverable: 5%
  - Mid-term demo: 10% (expert grading + popular opinion)
- World state replication: 10%
- Latency/jitter mitigation: 10%
- Final demo: 10% (expert grading + popular opinion)

# Bibliography

- Glazer, Josh, and Sanjay Madhav. **Multiplayer game programming: Architecting networked games**. Addison-Wesley Professional, 2015.
- Tanenbaum, Andrew S. **Computer Networks**. Pearson Education, 2013.





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# A brief history of networks and on-line games

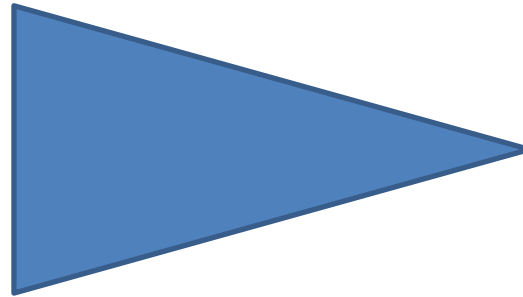


# History of computer networks

1961

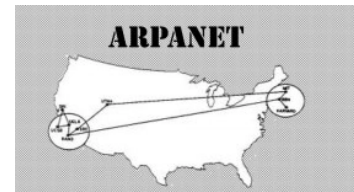
- Kleinrock (MIT) publishes theory about packet switching, which is essential for computer networking and having many concurrent communications.

Wikipedia



1966

- ARPAnet project is launched, aiming at creating a packet-switched computer network in the US.



# History of computer networks

1969

- For the first time, packets sent on a packet-switched network (ARPANET, only **4 nodes connected**).

1972

- The idea of an “open internetwork”, where different networks could be interconnected, was conceived together with the first protocols (later became TCP/IP)
  - Each network should stand on its own
  - Communications should be best effort
  - Simple black boxes would connect the different networks (those would become routers/gateways). Intelligence at the edge.
  - There is NO global control

1973

- First node from Europe added to the network with transatlantic connectivity. Ethernet is created at XEROX.

# History of computer networks

1974

- The first paper on TCP/IP is published. The term Internet was coined following the first steps from 1972.

1980

- Personal computers and Local Area Networks (LANs) start becoming a reality. It is demonstrated that TCP can be implemented in them and, thus, could be part of Internet.

1983

- ARPAnet finally migrates to TCP/IP, while DNS is invented to avoid remembering IP addresses. **60 nodes connected.**

# History of computer networks

1989

- Tim-Berners Lee creates the World Wide Web (WWW) at CERN. This includes the HyperText protocol, seed of the concept of “hyperlink”. **1000 nodes connected.**

1995

- Internet has become a widespread infrastructure with **50000+ nodes connected** in all continents.

2001

- 3G is deployed and internet in cell phones becomes relatively affordable. With 4G and the Internet of Things, this would lead to an explosion of nodes/data.

2011

- IPv4 addresses (over **4000M**) are exhausted

# History of on-line games

1958

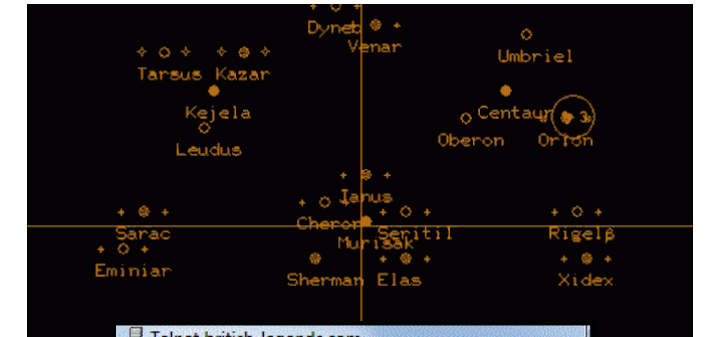
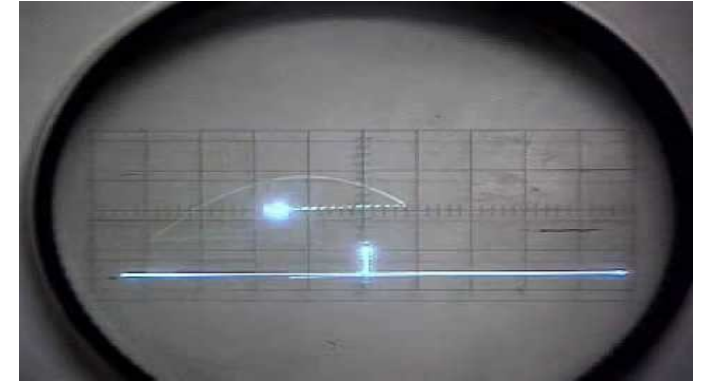
- First local multi-player games started appearing (Tennis for Two). They could be programmed as any other game, no need for connection.

1973

- First networked games appear for big mainframe computers, mostly primitive shooters and turn-based games (Empire, Maze War)

1978

- These games also inspired the rise of multi-user dungeons, usually text-based (Essex MUD)



# History of on-line games

1980

1981

1993

- As the university of Essex is connected to ARPANET, the game MUD becomes the first online game of the history.
- In parallel, personal computers and Local Area Networks (LANs) through the serial port start becoming a reality. First tutorials on how to program multiplayer games in this scope were released.
- The Internet starts to become widespread, but connections are still poor. LANs via Ethernet become very popular. LAN multiplayer starts to explode with games such as Doom.

<https://www.ionlitio.com/historia-id-software-doom/>





# History of on-line games

1996

- Internet starts to explode, speeds start becoming more decent. On-line gaming (through the Internet) takes flight, with games such as Quake or Unreal (1998) as first hits.



1998

- Massive on-line games also start becoming popular with widespread adoption of the Internet (Ultima Online)



2000

- Consoles join the party and start launching their services (SegaNet, Xbox Live, PlayStation Network) to allow on-line gaming in consoles as well.



# History of on-line games

2003

- Steam is launched and the on-line distribution of games becomes a viable and legal alternative.

2009

- Mobile games start gaining traction as 3G connectivity starts becoming affordable (Words with Friends).

2013

- Technologies evolve in cell phones and location-based and augmented reality games become widespread (Ingress, 2013)

2016

- Pokémon GO is launched and breaks all records. On-line videogame distribution is already a standard.

- On-line games proliferated along with the emergence of personal computers and networking.
- In on-line games, the main challenge is to have a consistent state across players in an efficient and fair way.
- The network is complex and unreliable and will make this illusion hard to obtain; but we'll give you the tools to try to make it happen.



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# Introduction

## Extra material

Sergi Abadal ([abadal@ac.upc.edu](mailto:abadal@ac.upc.edu))

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On-line games are distributed systems



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**ON-LINE GAMES  
ARE DISTRIBUTED  
SYSTEMS**

# What is a distributed system? (I)



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A distributed system is simply any environment where multiple computers or devices are working on a variety of tasks and components, all spread across a network. Components within distributed systems split up the work, coordinating efforts...

# What is a distributed system? (II)

## (From the perspective of the user)

A collection of (probably heterogeneous) processes whose distribution is transparent to the user so that the system appears as one local machine. This is in contrast to a network, where the user is aware that there are several machines, and their location, storage replication, load balancing and functionality is not transparent. Distributed systems **usually use some kind of client-server** organisation.



# What is a distributed system? (III)



CITM

**(From the perspective of the designer)**

A collection of processes, each of which is autonomous, programmable, asynchronous, and failure prone, and which communicate through an **unreliable communication medium**

- Examples of distributed systems
  - Client communicating with a server
  - The Internet
  - BitTorrent
  - Data centers
  - Dropbox
  - Google Drive
  - **On-line Games**